



OPEN Superclonal expansion and barley yellow dwarf virus transmission failure in invasive *Metopolophium festucae cerealium*

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Biological invasions can influence ecosystems, yet the evolutionary trajectories and disease dynamics of clonal expansion remain poorly understood. Since its 2011 detection, *Metopolophium festucae cerealium* (Stroyan), a subspecies of the cereal grass aphid endemic to Europe, has expanded across small grain growing regions of the western U.S. Our field surveys further indicate continued range expansion into previously unreported areas. Here, we integrate population genomics with virus transmission assays to evaluate its invasion structure and potential role in barley yellow dwarf virus (BYDV) transmission. Genome-wide SNP analysis of 262 individuals from six states reveals high genetic homogeneity, supporting a single introduction event and subsequent superclonal expansion. Although *M. f. cerealium* acquires BYDV at rates comparable to native *Rhopalosiphum padi* and maintains similar viral titers, it fails to transmit the virus. The primary agroecological risk posed by this invasion is through direct feeding injury, underscoring the need for continued monitoring and refined pest management strategies.

Keywords Biological invasion, Population genomics, Superclonal expansion, Vector competence, Barley yellow dwarf virus, Aphid, Cereal agroecosystems

Non-native insect invasions continue to threaten agroecosystems worldwide, largely a result of expanding global trade^{1–3}. Invasive species have the potential to establish rapidly in new environments with abundant hosts and few natural enemies, creating substantial agricultural, ecological, and economic challenges^{1–4}. In the U.S., invasive arthropod pests and pathogens are estimated to cause nearly \$40 billion in annual losses^{5,6}, and continued globalization is expected to increase the invasion risk⁶.

The cereal grass aphid, *Metopolophium festucae cerealium* Stroyan (Hemiptera: Aphididae), native to Western Europe and the British Isles^{7,8}, is an invasive pest species in western U.S. with the potential to threaten cereal production^{9–11}. *Metopolophium festucae cerealium* was first detected in the U.S. in 2011 in cultivated small grains of northern Oregon, Washington, and northern Idaho¹². Later surveys indicated a range expansion into cultivated and uncultivated grasses of eastern Idaho and southern Montana and Kansas, indicating its potential for becoming a widespread pest of cereals in the U.S.¹⁰.

In contrast to *M. festucae sensu lato*, which primarily infests perennial grasses within its native range, *M. f. cerealium* exploits a much wider host range of both perennial grasses and cereal crops in Europe and North America^{10,12,13}. In Europe, *M. f. cerealium* is primarily associated with wheat and, to a lesser extent, *Lolium* pasture grasses¹⁴. However, in the U.S., *M. f. cerealium* primarily colonizes wheat and barley, with the potential to persist on several uncultivated native grasses in the Poaceae family^{15,16}. Feeding injury by *M. f. cerealium* is substantial, reducing plant vigor and yield^{14,17}. In some cases, yield losses caused by *M. f. cerealium* feeding injury alone may exceed those caused by direct feeding of the bird cherry-oat aphid, *Rhopalosiphum padi* (L.) (Hemiptera: Aphididae), one of the most common cereal aphid species in the Pacific Northwest^{9,10}.

The epidemiological importance of *M. f. cerealium* stems from its potential role as a vector of yellow dwarf viruses (YDVs)¹⁸, one of the most damaging disease complexes of small grains¹⁹. Yellow dwarf disease is caused

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by a complex of aphid-transmitted barley and cereal yellow dwarf viruses (B/CYDVs), which are phloem-limited, positive-sense RNA viruses belonging primarily to the families *Tombusviridae* and *Solemoviridae*^{20,21}. This complex includes several economically important virus species, including BYDV-PAV, BYDV-PAS, BYDV-MAV, BYDV-kerII, BYDV-kerIII, CYDV-RPV, CYDV-RPS, maize yellow dwarf virus-RMV (MYDV-RMV), and other related yellow dwarf viruses such as wheat yellow dwarf virus (WYDV), barley virus G (BVG), and wheat leaf yellowing-associated virus (WLYaV)^{20,22–27}. These viruses infect a broad range of cultivated cereals and wild grasses and are transmitted in a persistent, circulative, non-propagative manner by more than 20 aphid species²⁸, which differ in their vector associations and transmission efficiencies^{29,30}. Barley yellow dwarf viruses can cause average yield losses of up to around 30%³¹ and, in rare cases, up to 100% in severe early infections³².

Several cereal aphids contribute to B/CYDV epidemiology, including *R. padi* (L.), *Sitobion avenae* (Fabricius), *Schizaphis graminum* (Rondani), *R. maidis* (Fitch), and *M. dirhodum* (Walker)^{30,33,34}. Among YDVs, BYDV-PAV (*Tombusviridae*; *Luteovirus pavhordei*) is the most prevalent and damaging species in the U.S. cereal production^{35,36}. *Rhopalosiphum padi* is considered one of the most important vectors because of its high transmission efficiency for BYDV-PAV^{36,37} and its widespread occurrence in cereal agroecosystems^{8,38}.

Although *M. festucae sensu stricto* is a documented vector of BYDV¹⁷, transmission by *M. f. cerealium* remains uncertain and previous laboratory evaluations did not detect successful BYDV-PAV transmission¹⁶. In Europe, *R. padi*, *S. avenae*, and *M. dirhodum* are considered the principal YDV vectors in cereal production systems³⁹. Other species within the genus *Metopolophium*, including *M. albidum*, *M. friscum*, and *M. festucae*, have also been reported as vectors of certain BYDV species⁴⁰. However, the vector status of *M. f. cerealium* remains uncertain. Previous studies involving *M. festucae* reported inconsistent transmission of BYDV, ranging from no transmission or very low transmission rates to successful transmission of certain BYDV species^{17,41–44}. Interpretation of these findings is complicated by historical taxonomic revisions because most studies did not distinguish between *M. festucae sensu stricto* and *M. f. cerealium*¹³. Consequently, the role of *M. f. cerealium* in YDV epidemiology remains unresolved.

Despite its expanding distribution and potential to inflict substantial injury to crops^{9–11}, the population genetic structure of *M. f. cerealium* across its invaded range, which could reflect the history and routes of its invasion, has not been assessed. Genetic structure and virus transmission by *M. f. cerealium* are linked because virus acquisition and transmission efficiency can vary among aphid genotypes and across geographic regions, especially in invasive aphid populations experiencing multiple introductions^{45–49}. Genetic variation can influence virulence, host use, and vector competence^{46,50,51}. Consequently, findings from a single population may not apply broadly if substantial spatial genetic variation exists. Population-level differences may further influence management-relevant traits such as economic injury levels and action thresholds. The risk of BYDV infection can be influenced by multiple interacting factors, including the abundance and distribution of aphid vectors, vector competence, virus species, host susceptibility, crop management practices, and environmental conditions⁵². Local aphid abundance and distribution, together with variation in host susceptibility and climatic conditions, may help identify areas at elevated risk of virus spread^{52,53}. In addition, information on aphid genotypes and their transmission efficiencies may improve predictions of disease incidence across agricultural landscapes^{49,54}. Integrating field surveys, population genomic analyses, and transmission assays can therefore improve risk assessment and clarify the role of invasive aphids in disease epidemiology.

Here, we integrated data based on field surveys, population genomics, and transmission bioassays to evaluate the invasion history and vector potential of *M. f. cerealium* in U.S. cereal agroecosystems. Field surveys conducted across multiple western states in 2022 and 2024 suggest ongoing geographic expansion, as *M. f. cerealium* was not detected in Wyoming and Utah in these surveys until 2024. We demonstrate that this species is spreading as a single superclone, consistent with a single introduction event. Unlike *M. festucae sensu stricto*, the *M. f. cerealium* subspecies does not appear to be a competent vector of BYDV (-PAV). The primary risk of the invasive *M. f. cerealium* is due to its direct feeding injury rather than virus transmission, refining our understanding of its ecological significance in agroecosystems.

Results

Geographic expansion of *M. f. cerealium* in the western U.S

During field surveys conducted across six states in 2022, *M. f. cerealium* was detected in Idaho, Oregon, Washington, and Montana, but not in Utah or Wyoming until 2024 (Supplementary Table 1; Fig. 1).

Population genetic structure reveals superclonal expansion of *M. f. cerealium*

Following library preparation and quality filtering, the final dataset comprised 18,519 SNPs from 288 individuals. PCA identified two major clusters clearly separating *M. f. cerealium* and *M. dirhodum*, with 10 individuals previously classified as *M. f. cerealium* clustering with *M. dirhodum* (Supplementary Fig. 1). Pairwise genetic distances corroborated this pattern, with the same 10 individuals clustering within the 16 previously identified *M. dirhodum* (Fig. 2). Nei's *Da* pairwise genetic distances between these 10 individuals and other *M. f. cerealium* all exceeded 0.8, while distances between these ten and the previously identified *M. dirhodum* were all less than 0.2. These 10 individuals were therefore reclassified as *M. dirhodum* and excluded from subsequent analyses.

A total of 262 *M. f. cerealium* individuals remained, and both PCA and Nei's *Da* pairwise genetic distances indicated minimal genetic variation (Supplementary Fig. 2). Of the 18,519 SNPs, only 1,454 were polymorphic within this group, and 93.8% (1,364) had a minor allele frequency less than 0.03, suggesting possible genotyping error. Most individuals showed a pairwise genetic distance of zero (Fig. 3), and further evidence suggested that these individuals are clones. Of the remaining 90 SNPs, 73 exhibited an observed heterozygote frequency greater than 0.9, whereas heterozygote frequencies greater than 0.5 are not expected under Hardy-Weinberg equilibrium. Because these loci are scattered across the genome, they are unlikely to result from misalignment of duplicate sequences; instead, they are most likely representing heterozygous loci inherited as a diploid genotype

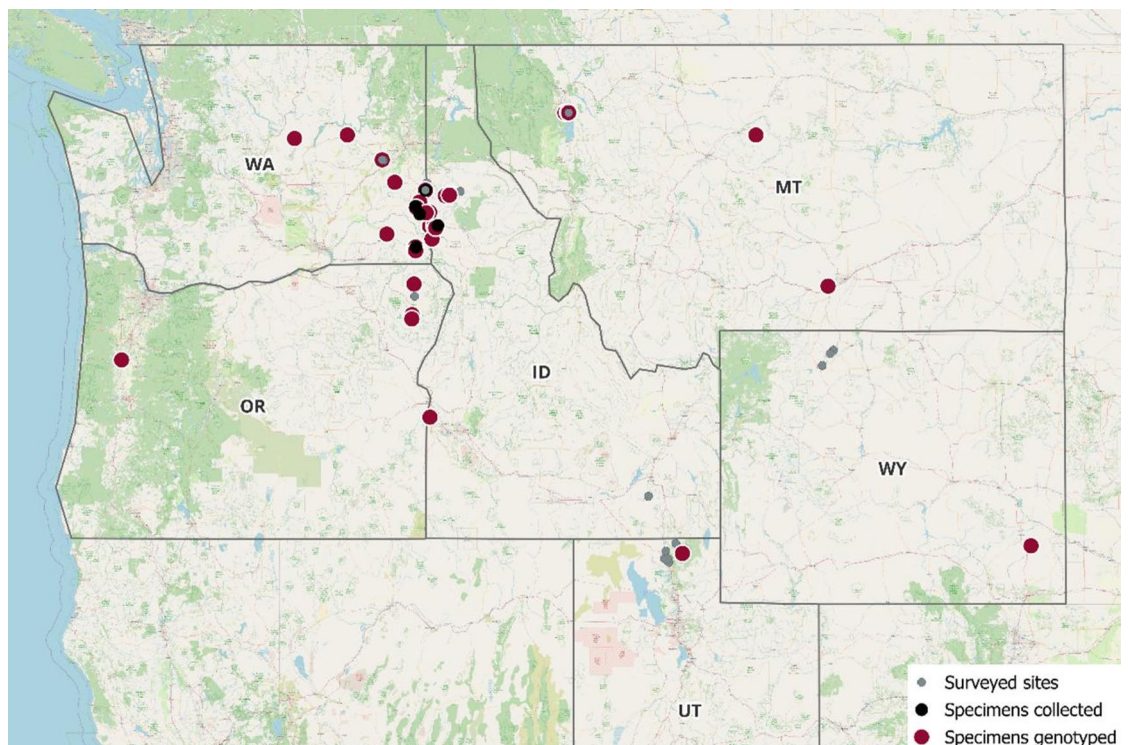


Fig. 1. Geographic distribution of sampling sites surveyed for the invasive cereal grass aphid *Metopolophium festucae cerealium* across the western U.S. Sampling locations are categorized based on sampling and downstream molecular analyses. Gray circles indicate surveyed sites, where locations were surveyed but no specimens were detected or genotyped. Black circles represent sites where aphid specimens were collected but not genotyped. Red circles denote sites where collected specimens were successfully genotyped.

through clonal reproduction. Consistent with this pattern, a Mantel test between genetic and geographic distance matrices detected no correlation ($r = 0.014$, $P = 0.184$), suggesting no evidence of isolation by distance.

We further investigated whether the SNPs with intermediate minor allele frequencies could represent new mutations within a subset of the clonal *M. f. cerealium* population. These SNPs were primarily found in a set of six individuals (Fig. 2). Heterozygous genotypes within these individuals consistently showed strong allelic imbalance, with the alternative allele represented at substantially lower read counts than the expected 50% (Supplementary Fig. 3). These individuals also showed higher levels of missing data at loci that appeared variable (Supplementary Fig. 4). These patterns indicate that these loci likely represent sequencing or PCR artifacts rather than true heterozygosity. While the six individuals are found in the spring cereals group, they are unlikely to reflect true genetic differentiation of this host group from other groups.

***Metopolophium festucae cerealium* acquires but does not transmit BYDV-PAV**

BYDV acquisition success did not differ between *R. padi* and *M. f. cerealium*, with 85% and 90% of individuals testing positive, respectively (Table 1). Similarly, virus titers were not significantly different between *R. padi* and *M. f. cerealium* (Table 1). However, despite comparable acquisition rates and the acquired BYDV titers, transmission outcomes differed markedly. Plants exposed to *M. f. cerealium* showed no successful BYDV inoculation (0%), whereas *R. padi* achieved a 60% inoculation success rate (Table 1).

Discussion

In this study, we integrated data based on field surveys, population genomic analyses, and virus transmission assays to evaluate the invasion dynamics and ecological relevance of *M. f. cerealium* in U.S. cereal agroecosystems. Field surveys conducted in 2022 and 2024 indicated likely geographic expansion, as *M. f. cerealium* was not detected in Wyoming and Utah in 2022 but was subsequently recorded in these states in 2024. Across extensive geographic sampling, we found no evidence of population genetic structure associated with region or host plant, indicating pronounced genetic homogeneity consistent with clonal expansion supporting a single introduction event. Despite its continued expansion and documented feeding damage^{14,17}, *M. f. cerealium* failed to transmit BYDV-PAV under controlled conditions, in contrast to the competent vector *R. padi*^{36,37}. Together, these results suggest that the agroecological impact of *M. f. cerealium* in its invaded range is driven primarily by direct feeding injury and indirect effects on aphid community dynamics, rather than by direct involvement in BYDV transmission.

Our genomic analyses found negligible variation among specimens sampled across the Pacific Northwest and Intermountain regions. The small number of variable loci within *M. f. cerealium* can be attributed either



Fig. 2. Pairwise genetic distance heatmap of *Metopolophium festucae cerealium* and *Metopolophium dirhodum*. Heatmap of Nei's D_a pairwise genetic distances of 18,519 SNPs between *M. f. cerealium* individuals and *M. dirhodum* outgroup individuals based on their pre-sequencing morphological identification as *M. f. cerealium* (MFC; red) or *M. dirhodum* (MD; blue). Genetic distances range from 0 (dark colors; high similarity) to 1 (light colors; high divergence). *M. dirhodum* individuals form a clearly distinct genetic cluster relative to *M. f. cerealium*, confirming their suitability as an outgroup and highlighting the extreme genetic uniformity within *M. f. cerealium* individuals. Notably, ten individuals initially identified as MFC cluster within the MD group, indicating probable misidentification prior to sequencing.

to sequencing error with extremely low apparent minor allele frequency, or heterozygous genotypes shared across all individuals. This latter pattern is inconsistent with genetic variation typically expected in an outbred population⁵⁵ and instead supports the dominance of a single multilocus genotype. These results indicate that the ongoing expansion of *M. f. cerealium* in the western U.S. represents a superclonal invasion, where populations across large geographic areas are dominated by a single multilocus genotype.

Such genetic uniformity is common among invasive aphids that reproduce primarily through obligate parthenogenesis, maintain only a few multilocus genotypes, and exploit a broad host range in both cultivated and uncultivated habitats^{56,57}. Asexual reproduction confers strong demographic advantages, which enable rapid population growth and range expansion even when genetic diversity is low^{58,59}. Similar superclonal dominance has been reported in other invasive aphid systems, including *Acyrtosiphon kondoi* Shinji in Australia^{58,60} and introductions of *R. padi* into South America^{61,62}, where one or a few multilocus genotypes occupy extensive geographic ranges.

Although sexual morphs likely occur within the native European range of *M. f. cerealium*^{8,13}, we detected no evidence of sexual reproduction in U.S. populations. While sexual reproduction cannot be definitively excluded, extensive surveys targeting putative overwintering hosts, including invasive and native rose species known to support sexual forms in related taxa, have failed to detect sexual populations (Wu & Eigenbrode, unpublished). The absence of host- or region-associated structure further supports the predominance of asexual reproduction following introduction. A comparable reduction in sexual reproduction has been documented in

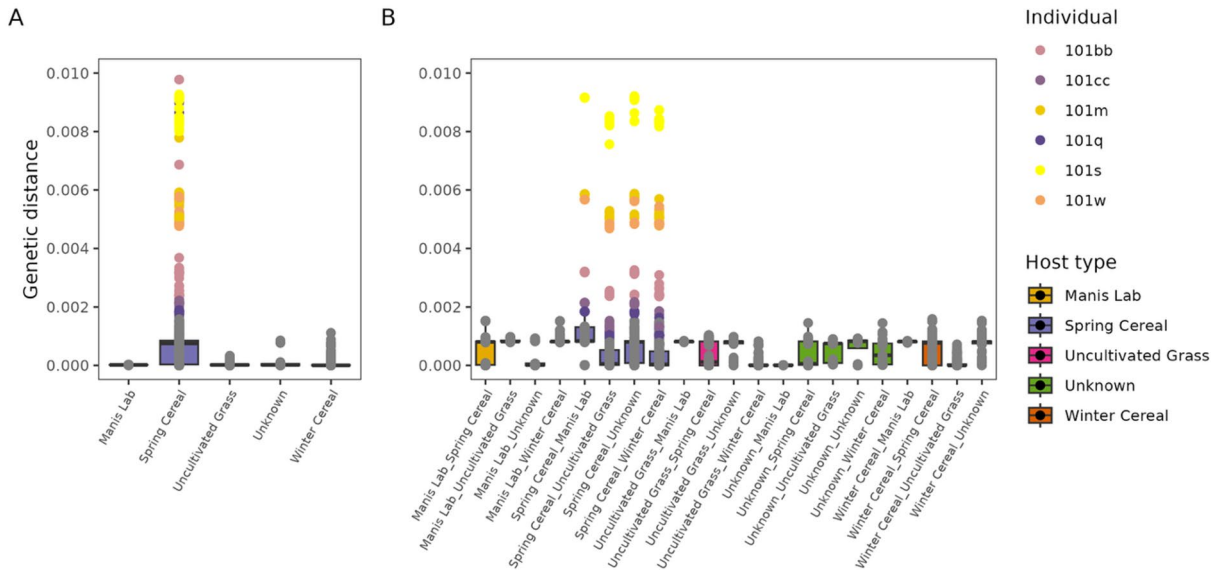


Fig. 3. Within- and between-host pairwise genetic distances in *Metopolophium festucae cerealium*. **(A)** Distribution of Nei's *Da* pairwise genetic distances among individuals collected within the same host category. **(B)** Distribution of pairwise genetic distances among individuals collected from different host categories. Host categories were defined based on whether aphids were collected from winter cereal crops, spring cereal crops, uncultivated grasses (non-agricultural habitats), maintained as a laboratory colony (Manis Lab), or from an unknown host. Colored points denote outlier individuals exhibiting greater genetic distances relative to the remaining 262 *M. f. cerealium* samples.

Aphid species	BYDV acquisition success (%) ¹	BYDV titer (mean ± SE copy numbers μL ⁻¹) ²	BYDV inoculation success (%)
<i>M. f. cerealium</i>	90	7,149.20 ± 1,793.54	0
<i>R. padi</i>	85	9,862.06 ± 2,870.48	60

Table 1. BYDV acquisition success, viral titer, and inoculation success in *Metopolophium festucae cerealium* and *Rhopalosiphum padi*. ¹BYDV acquisition success did not differ significantly between species (Wald $\chi^2_1 = 0.028$, $P = 0.867$). ²BYDV titers were not significantly different ($F_{1,36} = 0.166$, $P = 0.686$).

invasive U.S. populations of *M. dirhodum*, which are primarily anholocyclic relative to their holocyclic native counterparts^{8,63,64}.

Our genetic data provides a useful framework for interpreting transmission assays because vector genotype can influence virus transmission efficiency in aphid-virus systems, including BYDV complex⁴⁹. Comparable transmission variability has been reported among genetically distinct populations of *S. graminum* and genotypes of *R. padi*^{49,65}, whereas such effects were not detected in *S. avenae*⁶⁶. Because all populations examined here appear genetically similar, vector competence is expected to be relatively uniform across sampled populations. However, the introduction of genetically distinct lineages in the future could alter disease risk.

Both *M. f. cerealium* and *R. padi* acquired BYDV-PAV following feeding on infected plants; however, only *R. padi* successfully transmitted the virus to healthy plants, whereas *M. f. cerealium* failed to transmit BYDV-PAV. Although one might suggest that additional experimental repeats can further strengthen this conclusion, our current results are consistent with those of Sadeghi et al.¹⁶ and support that *M. f. cerealium* is not a competent vector of BYDV-PAV under the conditions tested. Furthermore, because transmission efficiency can vary among various combinations of BDYV and cereal aphid genotypes⁴⁸, our conclusions are limited to BYDV-PAV and the *M. f. cerealium* genotype evaluated in this study. Follow-up studies are needed to investigate the ability of *M. f. cerealium* to transmit other B/CYDVs and the related epidemiological significance¹⁶.

Transmission failure in *M. f. cerealium* was not associated with reduced virus titer, as BYDV titers did not differ significantly between these two aphid species. This suggests that barriers occur during internal virus movement rather than at the stage of acquisition. Persistent circulative transmission of luteoviruses requires receptor site-mediated movement through the midgut epithelium, circulate within the hemocoel, and reach the accessory salivary glands prior to inoculation^{45,67}. Disruption at any of these stages can prevent successful transmission. Differences in receptor compatibility, virion binding efficiency, or intracellular trafficking within aphid tissues may therefore explain the acquisition-without-transmission phenotype observed here⁶⁸.

Microbial symbionts may also influence transmission efficiency by modulating aphid immunity, gut physiology, or virion stability^{69,70}. For example, obligate symbionts such as *Buchnera aphidicola* produce

chaperonins, including GroEL, that can stabilize luteoviruses in the hemolymph and facilitate their persistence during circulative transmission^{65,71}. Recent studies suggest that facultative endosymbionts may also influence aphid-virus interactions through effects on aphid physiology, feeding behavior, and virus transmission, potentially contributing to variation in vector competence among aphid species, populations, or genotypes^{72,73}. Aphid digestive proteases, immune signaling pathways, and plant-derived metabolites can additionally influence virion uptake and circulation^{68,74–77}. Future studies are warranted to understand the underlying mechanism(s) rendering *M. f. cerealium* a non-vector of BYDV-PAV.

Vector competence within *Metopolophium* appears geographically variable. Certain populations of *M. dirhodum* and *M. festucae sensu stricto* transmit certain BYDV species in Europe and South America^{78,79}, whereas others do not³⁷. Recent studies from Australia and Europe also indicate that *M. dirhodum* can contribute to B/CYDV epidemiology, although transmission efficiency varies among virus species and isolates^{80,81}. Although our genomic analyses indicate low diversity among U.S. populations of *M. f. cerealium*, it is important that the transmission assays were conducted using a laboratory colony originally established in 2013 from individuals collected near Moscow, Idaho. Continued surveillance is warranted to detect any potential recent or future genotypes not captured in our surveys, which may differ in their virus-vector interactions.

Although we confirmed that *M. f. cerealium* is not a vector of BYDV-PAV, this aphid may still indirectly influence BYDV epidemiology within its invaded range. Non-vector species can alter aphid community dynamics, interspecific competition, or plant responses to aphid feeding⁸². Experimental studies demonstrate that *M. f. cerealium* enhances reproduction of *R. padi* on co-infested wheat plants^{10,83}, potentially increasing BYDV spread indirectly. Combined with its substantial per-capita feeding injury (Adhikari et al., unpublished) and broad host range across cultivated and uncultivated grasses^{10,13,15}, this species remains a significant management concern.

The superclonal structure documented here for *M. f. cerealium* extends the relevance of experimental findings across its invaded range. Provided its current population structure and absence of vectorial competence for BYDV-PAV, existing management strategies for *M. f. cerealium* are likely to be broadly applicable in the western U.S. However, the introduction of new genotypes can alter existing agroecological dynamics and/or BYDV epidemiology.

As a relatively new invasive species, there are several knowledge gaps that can be the focus of future studies. Examples include an understanding of *M. f. cerealium* vectorial capacity for other B/CYDV species, overwintering ecology, natural enemies and host plant resistance, as well as the identification of spectral signatures associated with its distinct feeding damage, all of which can inform risk assessment and adaptive IPM in our dynamic agroecosystems.

Methods

Population genetic structure of *M. f. cerealium*

Field collection of *M. f. cerealium*

Field surveys to detect *M. f. cerealium* were conducted in winter and spring cereal fields and uncultivated grasses across Idaho, Washington, Oregon, Montana, Utah, and Wyoming during 2022 and 2024 (Supplementary Table 1; Fig. 1), using the sweep netting method outlined by Halbert et al.¹². In brief, each site was systematically sampled using 150 standardized 180° sweeps along a zigzag transect (~100 m in length). Sampling started from the field edge and moved toward the center. Sampling began near the booting stage and continued from mid-June through August. Aphids collected from each location were placed in a single 2-mL microcentrifuge tube, preserved in 95% ethanol, and identified to species prior to downstream genetic analyses.

Specimen processing and DNA extraction

A total of 2,161 *M. f. cerealium* individuals were collected across six states during 2022 and 2024 (Supplementary Table 1). To ensure broad geographic representation while maintaining balanced sampling among sites, a subset of 314 individuals was selected for DNA extraction. Samples from sites with few individuals were retained whenever possible to maximize geographic coverage. The number of individuals per site varied (mean ± SD: 2.00 ± 5.01; median = 0; range = 0–33; Supplementary Table 1).

Our analysis also included five additional *M. f. cerealium* specimens from a laboratory colony originally established in 2013 from asexual virginiparous individuals collected in winter wheat fields near Moscow, Idaho, and maintained on barley at the Hubert C. Manis Entomological Laboratory (University of Idaho, Moscow, ID). These specimens were included for comparison with field-collected samples. In addition to the *M. f. cerealium* specimens, DNA from 16 rose grass aphids (*Metopolophium dirhodum* Walker) was extracted to be included as the outgroup in genetic analyses. DNA extractions were performed using liquid nitrogen and the DNeasy Blood & Tissue Kit (Qiagen, Hilden, Germany) following the manufacturer's protocol.

Of the 314 *M. f. cerealium* specimens selected for DNA extraction, three were excluded because DNA concentrations were too low for sequencing. In addition, two pooled samples were removed prior to single-nucleotide polymorphisms (SNPs) calling, resulting in a final dataset of 309 *M. f. cerealium* individuals for downstream analyses.

Library preparation

We used restriction-site associated DNA sequencing (RADseq; following the protocol of Ali et al.⁸⁴), a reduced-representation genomic sequencing technique to discover and genotype single-nucleotide polymorphisms (SNPs) across the genome. DNA samples from *M. f. cerealium* ($n = 309$) and *M. dirhodum* ($n = 16$) were quantified using a Qubit 2.0 Fluorometer (Thermo Fisher Scientific). DNA libraries were normalized to 5 ng/μL prior to SbfI enzyme digestion. Samples were ligated to barcoded adaptors and pooled prior to sonication to ~400 bp using a Covaris M220 Ultrasonicator at the Genomics and Bioinformatics Resources Core (GBRC; University of Idaho), followed by size selection from 200 to 700 bp. The NEBNext Ultra II DNA Library Prep Kit (E7645S) and

NEBNext i7 Multiplex Oligos (E7335S) were used to prepare libraries for Illumina sequencing according to the manufacturer's protocols. Fourteen samples were initially sequenced at the GBRC on half a lane of a MiSeq v2 (Nano, 150 bp paired-end sequencing), generating approximately 10 million reads per lane. Subsequent libraries were sequenced at the Genomics and Cell Characterization Core Facility (GC3F; University of Oregon) on two lanes of a NovaSeq 6000 SP flow cell, with 271 samples on the first lane and 40 samples on the second lane, generating approximately 400 million reads per lane.

Sample filtering and genotype calling

Libraries were demultiplexed by GC3F at the University of Oregon. Raw reads were filtered for the presence of a correct barcode and restriction enzyme cut site using a custom Perl script. Reads were then demultiplexed and filtered for PCR duplicates using the modules *process_radtags* and *clone-filter* in the software package Stacks (version 2.68; <https://catchenlab.life.illinois.edu/stacks/>)⁸⁵. Filtered reads were aligned to the *M. dirhodum* reference genome (GCA_019925205.1) using default parameters in the software package Burrows-Wheeler Aligner (version 0.7.17; <https://bio-bwa.sourceforge.net/>)⁸⁶. Aligned reads were filtered using the package SAMtools v1.13 (-F 2308) and sorted into BAM (Binary Alignment/Map) files⁸⁷. BAM files were then processed in Stacks v2.68 using the *ref_map.pl* pipeline with default parameters. SNPs were filtered using the *populations* module in Stacks v2.68, retaining loci with a minimum allele count of 3, a minimum genotype representation of 90% across individuals, and a minimum genotype depth of five reads. The resulting VCF (Variant Call Format) file was further filtered using VCFtools v0.1.16 to remove individuals with $\geq 10\%$ missing data⁸⁸.

Variant analysis

The filtered VCF file was used to generate a principal component analysis (PCA) in R v4.2.3 using the package adegenet v2.1.11 as an initial exploratory analysis to visualize genetic variation among individuals across both *M. f. cerealeum* and *M. dirhodum* as well as within *M. f. cerealeum* alone⁸⁹. Individual pairwise genetic distances were calculated using Nei's D_a in the *genet.dist* function of the R package hierfstat v0.5-11^{90,91}. The individual pairwise genetic distance matrix was visualized as a heatmap using the R package pheatmap v1.0.13, which by default clusters individuals hierarchically⁹². Geographic distances between sampling locations were also calculated as haversine distances using the R package geosphere v1.5-20⁹³. Genetic and geographic distances were compared using a Mantel test with Pearson's correlation coefficient and 999 permutations implemented in base R.

In addition to the 16 individuals collected and identified as *M. dirhodum*, 10 individuals initially classified as *M. f. cerealeum* exhibited very high genetic similarity to *M. dirhodum* and dissimilarity to the remaining *M. f. cerealeum* samples, as indicated by PCA and the pairwise genetic distance (see Results below). These 26 individuals were removed from subsequent analyses focusing on *M. f. cerealeum*. To identify loci contributing to genetic variation, a Mantel test was conducted between global pairwise genetic distances and distances calculated within each locus. PCA axis 1 was also used to identify loci contributing to observed variation. Sequencing depth and allele frequency were subsequently assessed to detect potential allele dropout at loci correlated with individual genetic variation, thereby evaluating the robustness of genotype calls (Mantel correlation filter).

BYDV vector competence assays

Plant and insect material

All plants and aphid colonies were maintained under controlled environmental conditions at the Hubert C. Manis Entomological Laboratory (University of Idaho, Moscow, ID). Wheat and *M. f. cerealeum* colonies were maintained in a greenhouse and *R. padi* colonies in a growth chamber, at 22–25 °C with a 16 h:8 h light:dark photoperiod. All transmission assays were conducted under the same environmental conditions in the greenhouse. Treatments were maintained in separate mesh cages (61 × 61 × 91 cm), with cages physically separated and spaced within the greenhouse to minimize aphid movement among cages.

Plants. Winter wheat (*Triticum aestivum* L., cv. 'SY-Ovation') was used in transmission assays. Individual seeds were planted in 10 × 10 × ~7.5 cm pots (~750 cm³) filled with a standardized soil mixture consisting of peatmoss (Sun Gro Horticulture Canada Ltd., Seba Beach, AB, Canada), sand, and fertilizer (Osmocote 20:20:20 N:P:K; Scott-Sierra Horticultural Products Co., Marysville, OH) at a 6:1:0.02 ratio. Wheat seedlings were used for transmission assay at the two-leaf stage, approximately two weeks post-germination.

Aphid colonies. The non-viruliferous *M. f. cerealeum* colony described in Sect. 1.2 was used for transmission assays. Both viruliferous and non-viruliferous colonies of *R. padi* were used as positive and negative controls, respectively, for BYDV-PAV (hereafter, BYDV) acquisition and transmission assays. *Rhopalosiphum padi* colonies were originally established from virginiparous specimens collected from small grain fields near Moscow, Idaho in 2001 and were reared on barley (*Hordeum vulgare* L., var. Sprinter). Viral infection status of both *R. padi* and *M. f. cerealeum* colonies was routinely monitored using reverse transcription–quantitative PCR (RT-qPCR; see below). The aphid colonies were not age-synchronized; however, only aphids of similar body size were used for experiments to minimize variation in feeding capacity and virus acquisition.

Virus source: The BYDV-PAV isolate used in this study originated from a viruliferous *R. padi* colony maintained on barley plants (var. Sprinter) infected with a Washington State isolate of BYDV-PAV¹⁶. Infection status was routinely confirmed using BYDV-PAV-specific RT-qPCR assays.

The overall design of the BYDV-PAV acquisition and transmission assay is illustrated in Fig. 1.

Generation of BYDV-infected source plants

Viruliferous *R. padi* were used to generate BYDV-infected source plants. Ten adult aphids of similar body size were caged on a single leaf (leaf stage 3 or 4) of each of 40 winter wheat plants using 2.54-cm foam clip cages with mesh ventilation (BDC1540, MegaView Science Co., Ltd., Taichung, Taiwan). After a 72-h inoculation access period (IAP), aphids were removed, and plants were maintained for 21 days to allow virus incubation before

being used as source plants for transmission assays. Plants were maintained in the greenhouse at 22–25 °C under a 16 h:8 h light: dark photoperiod. All source plants were tested for BYDV by RT-qPCR. No insecticides were applied to plants used in the inoculation bioassays, as these plants were subsequently used in the transmission experiments.

Acquisition and inoculation assays

Non-viruliferous *M. f. cerealium* and *R. padi* (10 adult aphids of similar body size per plant) were caged on a single leaf (leaf stage 3 or 4) of BYDV-infected source plants using 2.54 cm clip cages for a 72-h acquisition access period (AAP). Following acquisition, aphids were transferred to healthy two-week-old wheat seedlings (leaf stage 3–4; $N=20$ plants per species) for a 72-h IAP using clip cages.

Following the IAP, aphids were collected from the plants and stored at –80 °C for RT-qPCR testing to confirm virus acquisition. Plants were maintained for 21 days to allow viral incubation. After the incubation period, plants were tested for BYDV infection using RT-qPCR to determine transmission success. Control treatments included 20 wheat plants infested with non-viruliferous *R. padi*, 20 wheat plants infested with non-viruliferous *M. f. cerealium*, and 20 non-infested plants. All control plants tested negative for BYDV, confirming that no cross-contamination occurred; therefore, control plants were excluded from subsequent statistical analyses (see below).

Detection of BYDV infection

No visible BYDV symptoms were observed in infected plants during the experiment. BYDV infection in wheat and viruliferous status in aphids was assessed by RT-qPCR after 21 days, following Parizad and Rashed⁹⁴. Leaf tissue (approx. 100 mg) was collected from two fully expanded young leaves per plant. Total RNA was extracted using the RNeasy Plant Mini Kit (Qiagen, Hilden, Germany) and quantified using a NanoDrop spectrophotometer (NanoDrop Lite, Thermo Scientific, Madison, WI). RNA samples were subsequently normalized to 50 ng per 10 μ L reaction.

BYDV detection and copy number (titer) quantification were performed on a CFX Real-Time PCR System (Bio-Rad Laboratories, Hercules, CA, USA), using the Luna[®] Universal One-Step RT-qPCR Kit (New England Biolabs, Ipswich, MA). BYDV-PAV-specific primers described by Balaji et al.⁹⁵ were used. Each sample was tested in technical duplicates.

Reverse transcription was performed at 55 °C for 10 min. PCR amplification consisted of an initial denaturation at 95 °C for 1 min, followed by 40 cycles of 95 °C for 10 s and 60 °C for 1 min. To verify reaction specificity for absolute quantification, a melting-curve analysis was conducted by holding at 60 °C for 1 min and then increasing the temperature to 95 °C at a rate of 0.5 °C per cycle. A 10-fold serial dilution of the recombinant plasmid pIDTSmart Kan, containing a conserved fragment of the BYDV-PAV coat protein gene, was used to generate a standard curve. Viral copy numbers were calculated using the regression of C_q values against the log₁₀-transformed copy number.

Statistics and reproducibility

All statistical analyses were conducted using IBM SPSS Statistics version 29 (IBM Corp., Armonk, NY, USA) and R v4.2.3⁹⁶.

For genomic analyses, filtering thresholds were applied uniformly across all samples as described above. Individual aphids were treated as independent biological samples. Principal component analyses were performed without prior grouping assumptions. Isolation by distance was evaluated using a Mantel test with Pearson's correlation coefficient and 999 permutations. No data points were excluded from genomic analyses except those failing predefined filtering criteria ($\geq 10\%$ missing data or reclassification as *M. dirhodum* based on genetic clustering).

For transmission assays, virus acquisition success (viruliferous aphid = 1; non-viruliferous aphid = 0) and transmission success (infected plant = 1; uninfected plant = 0) were analyzed using binary logistic regression with aphid species as a fixed effect. Viral copy number (titer) in aphids was analyzed using a generalized linear mixed model with log₁₀-transformed copy numbers as the response variable. Model assumptions were evaluated using residual diagnostics and Shapiro-Wilk tests where appropriate.

Each wheat plant served as an independent biological replicate. Twenty plants per aphid species were used in acquisition and transmission assays. Aphids confined within a clip cage (10 individuals per plant) were treated as a single experimental unit for plant-level transmission analyses. All control plants tested negative for BYDV and were excluded from statistical comparisons.

Statistical significance was determined at $\alpha=0.05$. Data are presented as mean $\pm$ standard error of the mean (SEM) unless otherwise stated. Experiments were conducted under controlled environmental conditions as described.

Data availability

The raw sequencing data are available in the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA1443342 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1443342>). Source data underlying the figures are provided in Supplementary Table 1. All other data supporting the findings of this study are available within the manuscript and Supplementary Information or from the corresponding authors upon reasonable request.

Materials availability

Biological materials used in this study, including aphid colonies, are available from the corresponding authors

upon reasonable request, subject to institutional and phytosanitary regulations where applicable.

Code availability

This study used established software packages for sequence processing, variant calling, and statistical analysis, including Stacks v2.68, VCFtools v0.1.16, SAMtools v1.13, Burrows-Wheeler Aligner v0.7.17, IBM SPSS Statistics version 29, and R v4.2.3 with the packages adegenet v2.1.11, hierfstat v0.5-11, pheatmap v1.0.13, and geosphere v1.5-20. No custom code was generated that is central to the conclusions of this study.

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Author contributions

S.D.E., A.R. and P.H. conceived and funded the study. S.P., J.D., S.A. and M.F. conducted experiments and performed analyses. P.H. contributed to genomic analyses. A.R. and S.D.E. supervised the project. All authors contributed to writing and revising the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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